Variance of the Difference-in-Means Estimator for the SATE and Estimation of That Variance*

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This guide derives the variance of the Difference-in-Means estimator for the Sample Average Treatment Effect (SATE) under complete random assignment, and then develops a conservative estimator of that variance. The setup below states the framework only as far as the derivations require; the companion guide "Notation and Setup for Design-Based Causal Inference" gives the full development. These results extend to a superpopulation, where the estimand is the Population Average Treatment Effect (PATE); that extension is the companion guide "Variance of the Difference-in-Means Estimator for the PATE and Estimation of That Variance".

1 Variance of the Difference-in-Means estimator for the Sample Average Treatment Effect (SATE)

1.1 Setup

This section states the framework only as far as the derivations require; the companion guide "Notation and Setup for Design-Based Causal Inference" develops each object in full, with the reasoning behind it.

Let the index $i \in \{1, \dots, n\}$ run over the n units of a finite sample, $\mathcal{S}_n$, of which $n_T \geq 2$ receive treatment and $n_C = n - n_T \geq 2$ receive control, so that $n \geq 4$; these bounds ensure that the variance estimator developed below, which forms a separate sample variance within each arm,

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is well defined. The indicator $Z_i \in \{0,1\}$ records whether unit i receives treatment ($Z_i = 1$) or control ($Z_i = 0$), and the random assignment vector $\boldsymbol{Z} = \begin{bmatrix} Z_1, \ldots, Z_n \end{bmatrix}^\top$ collects the indicators. Write $n_T(\boldsymbol{z}) \coloneqq \sum_{i=1}^n z_i$ for the number of treated units under an assignment $\boldsymbol{z}$, so that $n_T(\boldsymbol{Z})$ is a random variable. Under *complete random assignment* the design fixes the count: The support of $\boldsymbol{Z}$ is $\Omega = \{\boldsymbol{z} \in \{0,1\}^n : n_T(\boldsymbol{z}) = n_T\}$, on which $n_T(\boldsymbol{Z})$ equals the design constant n_T with probability one, and $\boldsymbol{Z}$ is distributed uniformly on Ω, whose cardinality[1] is $|\Omega| = \binom{n}{n_T}$. The unbiasedness guide shows how the same analysis covers simple random assignment after conditioning on the event $\{n_T(\boldsymbol{Z}) = n_T\}$.

Under the Stable Unit Treatment Value Assumption (SUTVA) (Cox, 1958; Rubin, 1980, 1986), unit i's potential outcome depends on the assignment vector only through unit i's own coordinate, so each unit has two *fixed* potential outcomes, $y_i(1)$ and $y_i(0)$, with individual effect $\tau_i \coloneqq y_i(1) - y_i(0)$. The observable outcome is $Y_i \coloneqq y_i(Z_i) = Z_i y_i(1) + (1 - Z_i) y_i(0)$, random only through Z_i. The setup guide develops the potential outcomes schedule, states SUTVA precisely, and walks through the collapse from $y_i(\boldsymbol{Z})$ to $y_i(1)$ and $y_i(0)$ in numbered steps.

Write $\bar{y}(1) \coloneqq n^{-1} \sum_{i=1}^n y_i(1)$ and $\bar{y}(0) \coloneqq n^{-1} \sum_{i=1}^n y_i(0)$ for the finite population means of the treatment and control potential outcomes; both are fixed quantities. The target of interest is the Sample Average Treatment Effect (SATE),

$$\tau_{\text{SATE}} \coloneqq n^{-1} \sum_{i=1}^n \tau_i = \bar{y}(1) - \bar{y}(0),$$

which is likewise a fixed (though unknown) quantity. Define the Difference-in-Means estimator of τ_{SATE} under complete random assignment as

$$\hat{\tau} \coloneqq \frac{1}{n_T} \sum_{i=1}^n Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^n (1 - Z_i) Y_i. \tag{1}$$

The estimator in Equation (1) is the same Difference-in-Means estimator whose unbiasedness the companion guide establishes. Under complete random assignment, where the design fixes the counts $n_T(\boldsymbol{Z}) = n_T$ and $n_C(\boldsymbol{Z}) = n_C$, the estimator coincides with the ratio form $(\sum_i Z_i Y_i)/(\sum_i Z_i) - (\sum_i (1 - Z_i) Y_i)/(\sum_i (1 - Z_i))$. Unlike the estimand τ_{SATE}, the estimator $\hat{\tau}$ is a *random variable*, since $\hat{\tau}$ is a function of the random assignment vector $\boldsymbol{Z}$ and inherits all randomness from $\boldsymbol{Z}$. For the expectation and variance of $\hat{\tau}$ in Equation (1), I write $\mathrm{E}_\Omega[\cdot]$ and $\mathrm{Var}_\Omega[\cdot]$ to emphasize that they run over only the randomness of the assignment process, i.e., over the distribution of $\boldsymbol{Z}$ on Ω.

[1]For an arbitrary set W, let $|W|$ denote the cardinality of (i.e., the number of elements in) the set W.

1.2 Derivation of variance of Difference-in-Means estimator for the SATE

Proposition 1. *The variance of $\hat{\tau}$ for the τ_{SATE} under complete random assignment is*

$$\operatorname{Var}_\Omega[\hat{\tau}] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}, \tag{2}$$

where

$$S_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1)\right)^2 \tag{3}$$

$$S_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0)\right)^2 \tag{4}$$

$$S_n^2(\boldsymbol{\tau}) = \left(\frac{1}{n-1}\right)\sum_{i=1}^{n}\left(\tau_i - \frac{1}{n}\sum_{i=1}^{n} \tau_i\right)^2. \tag{5}$$

Proof. Building on Imbens and Rubin (2015, Chapter 6, Appendix A), I will break the proof into several steps:

Step 1: I will show that we can rewrite the Difference-in-Means estimator in (1) in terms of *not* the raw assignment indicators Z_i, but the centered treatment variable $A_i := Z_i - \mathrm{E}_\Omega[Z_i]$. I center because the variance measures the fluctuations of $\hat{\tau}$ around its mean, τ_{SATE}. If we can algebraically separate $\hat{\tau}$ into a fixed part (equal to τ_{SATE}) plus a random part with mean zero, then the variance of $\hat{\tau}$ equals the variance of the random part alone. Centering the assignment variable produces this separation and, because $\mathrm{E}_\Omega[A_i] = 0$, greatly simplifies the algebra that follows.

Step 2: The potential outcomes, $\{y_i(1), y_i(0)\}_{i=1}^n$, are fixed, and the observable outcomes inherit their randomness only from the treatment assignment. Once I write $\hat{\tau}$ as a weighted sum of the A_i, its variance depends only on the first and second moments of the A_i. I will therefore derive $\mathrm{E}_\Omega[A_i]$, $\operatorname{Var}_\Omega[A_i]$ and $\operatorname{Cov}[A_i, A_j]$ for $i \neq j$. The covariance term is essential and nonzero. Under complete random assignment the design fixes the count $n_T(\boldsymbol{Z})$ at n_T, so if one unit is pulled into treatment another must be pulled out. This mechanical dependence among the assignment indicators is the source of the covariance term, and the covariance turns out to be negative.

Step 3: I will rely on the expressions for $S_n^2(\boldsymbol{y}(1))$ and $S_n^2(\boldsymbol{y}(0))$ above, as well as on the following algebraic equivalence that I will need to prove:

$$S_n^2(\boldsymbol{\tau}) = S_n^2(\boldsymbol{y}(1)) + S_n^2(\boldsymbol{y}(0)) - \frac{2}{n-1}\sum_{i=1}^{n}\left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1)\right)\left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0)\right). \tag{6}$$

Step 4: Once I have completed the three previous steps, I will rely only on the linearity of expectations, rules of variance and (often very messy) algebra.

Step 1

First, we can re-write the estimator as

$$\frac{1}{n_T}\sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_C}\sum_{i=1}^{n}(1-Z_i)\,Y_i = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{n}{n_T}Z_i y_i(1) - \frac{n}{n_C}(1-Z_i)\,y_i(0)\right)$$

Because $n = n_C + n_T$, $1 = \frac{n_C+n_T}{n}$, so instead of $(1-Z_i)$ we can write $(\frac{n_C+n_T}{n} - Z_i)$:

$$\frac{1}{n}\sum_{i=1}^{n}\frac{n}{n_T}Z_i y_i(1) - \frac{n}{n_C}\left(\frac{n_C+n_T}{n} - Z_i\right) y_i(0),$$

which after a little bit of algebra yields

$$\frac{1}{n}\sum_{i=1}^{n}\frac{n}{n_T}\left(\underbrace{Z_i - \frac{n_T}{n}}_{=A_i} + \frac{n_T}{n}\right) y_i(1) - \frac{n}{n_C}\left[\frac{n_C}{n} - \left(\underbrace{Z_i - \frac{n_T}{n}}_{=A_i}\right)\right] y_i(0), \tag{7}$$

where A_i is the centered treatment variable, $A_i = Z_i - \mathrm{E}_\Omega[Z_i]$, because, under complete random assignment, $\mathrm{E}_\Omega[Z_i] = \frac{n_T}{n}$.

Therefore, we can re-write (7) as

$$\frac{1}{n}\sum_{i=1}^{n}\frac{n}{n_T}\left(A_i + \frac{n_T}{n}\right) y_i(1) - \frac{n}{n_C}\left(\frac{n_C}{n} - A_i\right) y_i(0),$$

which, with a bit more algebra, is equivalent to

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}\big(y_i(1) - y_i(0)\big)}_{=\tau_{\text{SATE}}} + \frac{1}{n}\sum_{i=1}^{n} A_i\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right) = \tau_{\text{SATE}} + \frac{1}{n}\sum_{i=1}^{n} A_i\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)$$

Step 2

Thus far, I have shown that the Difference-in-Means estimator in (1) is equivalent to

$$\tau_{\text{SATE}} + \frac{1}{n}\sum_{i=1}^{n} A_i \left(\frac{n}{n_T} y_i(1) + \frac{n}{n_C} y_i(0)\right), \tag{8}$$

in which τ_{SATE}, n_C, n_T, n, and $\{y_i(0), y_i(1)\}_{i=1}^n$ are all fixed quantities. The only random quantities in (8) are $\{A_i\}_{i=1}^n$.

Therefore, to derive the variance of (8), I now need to derive $\mathrm{E}_\Omega[A_i]$, $\mathrm{Var}_\Omega[A_i]$ and $\mathrm{E}_\Omega[A_i A_j]$ for $i \neq j$.

To do so, I write the sample space of A_i:

$$A_i = Z_i - \frac{n_T}{n} = \begin{cases} \dfrac{n_C}{n} & \text{if } Z_i = 1 \\ -\dfrac{n_T}{n} & \text{if } Z_i = 0 \end{cases} \tag{9}$$

and recall that, under complete random assignment, $\Pr_\Omega(Z_i = 1) = \frac{n_T}{n}$ and $\Pr_\Omega(Z_i = 0) = 1 - \frac{n_T}{n} = \frac{n}{n} - \frac{n_T}{n} = \frac{n - n_T}{n} = \frac{n_C}{n}$.

Derivation of $\mathrm{E}_\Omega[A_i]$**:** Substituting the definition of A_i gives $\mathrm{E}_\Omega[A_i] = \mathrm{E}_\Omega[Z_i - \frac{n_T}{n}] = \mathrm{E}_\Omega[Z_i] - \frac{n_T}{n} = \frac{n_T}{n} - \frac{n_T}{n} = 0$ or, equivalently,

$$\begin{aligned}
\mathrm{E}_\Omega\left[A_i\right] &= \frac{n_C}{n} \Pr_\Omega\left(Z_i = 1\right) - \frac{n_T}{n} \Pr_\Omega\left(Z_i = 0\right) \\
&= \frac{n_C}{n}\frac{n_T}{n} - \frac{n_T}{n}\left(1 - \frac{n_T}{n}\right) \\
&= \frac{n_C}{n}\frac{n_T}{n} - \frac{n_T}{n}\frac{n_C}{n} \\
&= 0.
\end{aligned}$$

Derivation of $\mathrm{Var}_\Omega[A_i]$**:** Since $\mathrm{E}_\Omega[A_i] = 0$, the variance reduces to $\mathrm{Var}_\Omega[A_i] = \mathrm{E}_\Omega[A_i^2] - \mathrm{E}_\Omega[A_i]^2 = \mathrm{E}_\Omega[A_i^2]$. Thus,

$$\begin{aligned}
\mathrm{Var}_\Omega\left[A_i\right] &= \mathrm{E}_\Omega\left[A_i^2\right] \\
&= \left(\frac{n_C}{n}\right)^2 \Pr_\Omega\left(Z_i = 1\right) + \left(-\frac{n_T}{n}\right)^2 \Pr_\Omega\left(Z_i = 0\right) \\
&= \frac{n_C^2}{n^2} \Pr_\Omega\left(Z_i = 1\right) + \frac{n_T^2}{n^2} \Pr_\Omega\left(Z_i = 0\right)
\end{aligned}$$

$$
\begin{aligned}
&= \frac{n_C^2}{n^2}\frac{n_T}{n} + \frac{n_T^2}{n^2}\frac{n_C}{n} \\
&= \frac{n_C^2 n_T}{n^3} + \frac{n_T^2 n_C}{n^3} \\
&= \frac{n_C^2 n_T + n_T^2 n_C}{n^3} \\
&= \frac{n_C n_T \left(n_C + n_T\right)}{n^3} \\
&= \frac{n_C n_T \left(n\right)}{n^3} \\
&= \frac{n_C n_T}{n^2}.
\end{aligned}
$$

Derivation of $\text{Cov}[A_i, A_j]$ **for** $i \neq j$**:** The $\{A_i\}_{i=1}^{n}$ all have the same expected value, $\text{E}_\Omega[A_i] = 0$ (as I just showed), but they are *not* independent: Fixing the count $n_T(\boldsymbol{Z})$ at n_T links the assignment indicators to one another. I therefore will need to derive $\text{Cov}[A_i, A_j]$ for $i \neq j$. To do so, first note that

$$
\begin{aligned}
\text{Cov}\left[A_i, A_j\right] &= \text{E}_\Omega\left[\left(A_i - \text{E}_\Omega\left[A_i\right]\right)\left(A_j - \text{E}_\Omega\left[A_j\right]\right)\right] \\
&= \text{E}_\Omega\left[\left(A_i - 0\right)\left(A_j - 0\right)\right] \\
&= \text{E}_\Omega\left[A_i A_j\right],
\end{aligned}
$$

which is easier to work with.

The possible values that $A_i A_j$ could take on are

$$
A_i A_j = \begin{cases}
\left(\frac{n_C}{n}\right)\left(\frac{n_C}{n}\right) = \frac{n_C^2}{n^2} & \text{if } Z_i = 1 \text{ and } Z_j = 1 \\
\left(-\frac{n_T}{n}\right)\left(\frac{n_C}{n}\right) = \frac{-n_T n_C}{n^2} & \text{if } Z_i = 0 \text{ and } Z_j = 1 \text{ or } Z_i = 1 \text{ and } Z_j = 0 \\
\left(-\frac{n_T}{n}\right)\left(-\frac{n_T}{n}\right) = \frac{n_T^2}{n^2} & \text{if } Z_i = 0 \text{ and } Z_j = 0.
\end{cases}
$$

Having derived the sample space of $A_i A_j$, I now need to derive the probabilities that correspond to each of the events in that sample space, namely, $\text{Pr}_\Omega(Z_i = 1, Z_j = 1)$, $\text{Pr}_\Omega(Z_i = 0, Z_j = 1)$, $\text{Pr}_\Omega(Z_i = 1, Z_j = 0)$ and $\text{Pr}_\Omega(Z_i = 0, Z_j = 0)$.

Remember that Z_i and Z_j are *not* independent, which implies that, e.g., $\text{Pr}_\Omega(Z_i = 1, Z_j = 1) \neq \text{Pr}_\Omega(Z_i = 1)\,\text{Pr}_\Omega(Z_j = 1)$. Instead, I will appeal to the definition of joint probability in which

$\Pr_\Omega(Z_i = z, Z_j = z') = \Pr_\Omega(Z_i = z) \Pr_\Omega(Z_j = z' \mid Z_i = z)$:

$$\Pr_\Omega\left(Z_i = z, Z_j = z'\right) = \begin{cases} \left(\frac{n_T}{n}\right)\left(\frac{n_T - 1}{n-1}\right) & \text{if } Z_i = 1 \text{ and } Z_j = 1 \\ \left(\frac{n_T}{n}\right)\left(\frac{n_C}{n-1}\right) & \text{if } Z_i = 1 \text{ and } Z_j = 0 \\ \left(\frac{n_C}{n}\right)\left(\frac{n_T}{n-1}\right) & \text{if } Z_i = 0 \text{ and } Z_j = 1 \\ \left(\frac{n_C}{n}\right)\left(\frac{n_C}{n-1}\right) & \text{if } Z_i = 0 \text{ and } Z_j = 0. \end{cases}$$

We can therefore write the probability mass function (PMF) of $A_i A_j$ as

$$\Pr_\Omega\left(A_i A_j\right) = \begin{cases} \left(\frac{n_T}{n}\right)\left(\frac{n_T - 1}{n-1}\right) = \frac{n_T\left(n_T - 1\right)}{n\left(n-1\right)} & \text{if } A_i A_j = \frac{n_C^2}{n^2} \\ \left(\frac{n_T}{n}\right)\left(\frac{n_C}{n-1}\right) + \left(\frac{n_C}{n}\right)\left(\frac{n_T}{n-1}\right) = \frac{2\left(n_T n_C\right)}{n\left(n-1\right)} & \text{if } A_i A_j = \frac{-n_T n_C}{n^2} \\ \left(\frac{n_C}{n}\right)\left(\frac{n_C}{n-1}\right) = \frac{n_C\left(n_C - 1\right)}{n\left(n-1\right)} & \text{if } A_i A_j = \frac{n_T^2}{n^2}, \end{cases}$$

which implies that

$$\mathrm{E}_\Omega\left[A_i A_j\right] = \frac{n_C^2}{n^2}\left(\frac{n_T\left(n_T - 1\right)}{n\left(n-1\right)}\right) + \frac{-n_T n_C}{n^2}\left(\frac{2\left(n_T n_C\right)}{n\left(n-1\right)}\right) + \frac{n_T^2}{n^2}\left(\frac{n_C\left(n_C - 1\right)}{n\left(n-1\right)}\right)$$

or equivalently, after some algebra,

$$\frac{-n_T n_C}{n^2\left(n-1\right)}.$$

Step 3

In this step, I will prove the algebraic equivalence between the expression for $S_n^2(\boldsymbol{\tau})$ in Equation (5) above and

$$S_n^2(\boldsymbol{\tau}) = S_n^2(\boldsymbol{y}(1)) + S_n^2(\boldsymbol{y}(0)) - \frac{2}{n-1}\sum_{i=1}^{n}\left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1)\right)\left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0)\right), \tag{10}$$

which will be valuable for the derivation of $\mathrm{Var}_\Omega[\hat{\tau}]$ in **Step 4** to follow.

Recall from Equation (5) that

$$S_n^2(\boldsymbol{\tau}) = \frac{1}{n-1}\sum_{i=1}^{n}\left(\tau_i - \frac{1}{n}\sum_{i=1}^{n}\tau_i\right)^2.$$

Hence,

$$
\begin{aligned}
S_n^2(\boldsymbol{\tau}) &= \frac{1}{n-1}\sum_{i=1}^{n}\left(\tau_i - \frac{1}{n}\sum_{i=1}^{n}\tau_i\right)^2 \\
&= \frac{1}{n-1}\sum_{i=1}^{n}\left(\underbrace{\left(y_i(1)-y_i(0)\right)}_{=\tau_i} - \frac{1}{n}\sum_{i=1}^{n}\underbrace{\left(y_i(1)-y_i(0)\right)}_{=\tau_i}\right)^2 \\
&= \frac{1}{n-1}\sum_{i=1}^{n}\left(\left(y_i(1)-\frac{1}{n}\sum_{i=1}^{n}y_i(1)\right) - \left(y_i(0)-\frac{1}{n}\sum_{i=1}^{n}y_i(0)\right)\right)^2,
\end{aligned}
$$

where, in the last line, I have used $\frac{1}{n}\sum_{i=1}^{n}(y_i(1)-y_i(0)) = \frac{1}{n}\sum_{i=1}^{n}y_i(1) - \frac{1}{n}\sum_{i=1}^{n}y_i(0)$ to regroup each summand into a treated deviation, $(y_i(1)-\frac{1}{n}\sum_{i=1}^{n}y_i(1))$, minus a control deviation, $(y_i(0)-\frac{1}{n}\sum_{i=1}^{n}y_i(0))$. Expanding the square via $(a-b)^2 = a^2 - 2ab + b^2$ and then distributing the sum yields

$$
\begin{aligned}
S_n^2(\boldsymbol{\tau}) &= \underbrace{\frac{1}{n-1}\sum_{i=1}^{n}\left(y_i(1)-\frac{1}{n}\sum_{i=1}^{n}y_i(1)\right)^2}_{=S_n^2(\boldsymbol{y}(1))} + \underbrace{\frac{1}{n-1}\sum_{i=1}^{n}\left(y_i(0)-\frac{1}{n}\sum_{i=1}^{n}y_i(0)\right)^2}_{=S_n^2(\boldsymbol{y}(0))} \\
&\quad - \frac{2}{n-1}\sum_{i=1}^{n}\left(y_i(1)-\frac{1}{n}\sum_{i=1}^{n}y_i(1)\right)\left(y_i(0)-\frac{1}{n}\sum_{i=1}^{n}y_i(0)\right) \\
&= S_n^2(\boldsymbol{y}(1)) + S_n^2(\boldsymbol{y}(0)) - \frac{2}{n-1}\sum_{i=1}^{n}\left(y_i(1)-\frac{1}{n}\sum_{i=1}^{n}y_i(1)\right)\left(y_i(0)-\frac{1}{n}\sum_{i=1}^{n}y_i(0)\right).
\end{aligned}
$$

Step 4

To complete the derivation, I compute the variance of the Difference-in-Means estimator from the expression in (8):

$$
\begin{aligned}
\mathrm{Var}_\Omega\left[\hat{\tau}\right] &= \mathrm{Var}_\Omega\left[\tau_{\text{SATE}} + \frac{1}{n}\sum_{i=1}^{n}A_i\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\right] \\
&= \frac{1}{n^2}\,\mathrm{Var}_\Omega\left[\sum_{i=1}^{n}A_i\underbrace{\left(\frac{n}{n_T}y_i(1)+\frac{n}{n_C}y_i(0)\right)}_{\text{constant}}\right],
\end{aligned}
$$

which, since $\mathrm{E}_\Omega[A_i] = 0$, is equivalent to

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{1}{n^2}\left(\mathrm{E}_\Omega\left[\left(\sum_{i=1}^{n} A_i \underbrace{\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)}_{\text{constant}}\right)^2\right] - \mathrm{E}_\Omega\left[\sum_{i=1}^{n} A_i \underbrace{\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)}_{\text{constant}}\right]^2\right)$$

$$= \frac{1}{n^2}\,\mathrm{E}_\Omega\left[\left(\sum_{i=1}^{n} A_i \left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\right)^2\right].$$

Expanding the expression immediately above yields

$$\begin{aligned}
\mathrm{Var}_\Omega\left[\hat{\tau}\right] &= \frac{1}{n^2}\,\mathrm{E}_\Omega\left[\left(\sum_{i=1}^{n} A_i \left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\right)^2\right] \\
&= \frac{1}{n^2}\,\mathrm{E}_\Omega\left[\left(\sum_{i=1}^{n} A_i \left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\right)\left(\sum_{j=1}^{n} A_j \left(\frac{n}{n_T}y_j(1) + \frac{n}{n_C}y_j(0)\right)\right)\right] \\
&= \frac{1}{n^2}\,\mathrm{E}_\Omega\left[\sum_{i=1}^{n}\sum_{j=1}^{n} A_i A_j \left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\left(\frac{n}{n_T}y_j(1) + \frac{n}{n_C}y_j(0)\right)\right].
\end{aligned}$$

In **Step 2** above, I showed that

$$\begin{cases}
\mathrm{E}_\Omega\left[A_i A_j\right] = \dfrac{-n_T n_C}{n^2\,(n-1)} & \text{when } i \neq j \\
\mathrm{E}_\Omega\left[A_i A_j\right] = \mathrm{E}_\Omega\left[A_i^2\right] = \mathrm{Var}_\Omega\left[A_i\right] = \dfrac{n_C n_T}{n^2} & \text{when } i = j.
\end{cases}$$

I now split the double sum $\sum_{i=1}^{n}\sum_{j=1}^{n}$ into the n diagonal terms (where $i = j$, so that $A_i A_j = A_i^2$) and the $n(n-1)$ off-diagonal terms (where $i \neq j$), because $\mathrm{E}_\Omega[A_i A_j]$ takes a different value in each case. The split yields

$$\begin{aligned}
\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{1}{n^2}\Bigg(&\sum_{i=1}^{n}\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)^2 \underbrace{\mathrm{E}_\Omega\left[A_i^2\right]}_{=\frac{n_C n_T}{n^2}} \\
&+ \sum_{i=1}^{n}\sum_{j\neq i}\left(\frac{n}{n_T}y_i(1) + \frac{n}{n_C}y_i(0)\right)\left(\frac{n}{n_T}y_j(1) + \frac{n}{n_C}y_j(0)\right)\underbrace{\mathrm{E}_\Omega\left[A_i A_j\right]}_{=\frac{-n_T n_C}{n^2\,(n-1)}}\Bigg).
\end{aligned}$$

I now substitute the expressions I derived for $\mathrm{E}_\Omega[A_i^2]$ and $\mathrm{E}_\Omega[A_i A_j]$. To keep the bookkeeping manageable, I abbreviate the constant weight multiplying A_i as $w_i := \frac{n}{n_T} y_i(1) + \frac{n}{n_C} y_i(0)$, so that the expression above becomes

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{1}{n^2}\left(\frac{n_C n_T}{n^2}\sum_{i=1}^n w_i^2 - \frac{n_T n_C}{n^2\left(n-1\right)}\sum_{i=1}^n\sum_{j\neq i} w_i w_j\right).$$

Using the identity $\sum_{i=1}^n \sum_{j\neq i} w_i w_j = (\sum_{i=1}^n w_i)^2 - \sum_{i=1}^n w_i^2$ and then collecting terms over the common denominator $n-1$,

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{n_C n_T}{n^4\left(n-1\right)}\left(n\sum_{i=1}^n w_i^2 - \left(\sum_{i=1}^n w_i\right)^2\right) = \frac{n_C n_T}{n^3\left(n-1\right)}\sum_{i=1}^n \left(w_i - \bar{w}\right)^2,$$

where $\bar{w} = \frac{1}{n}\sum_{i=1}^n w_i$, and where the final equality uses the identity $n\sum_i w_i^2 - (\sum_i w_i)^2 = n\sum_i (w_i - \bar{w})^2$. Finally, I substitute back the centered weight,

$$w_i - \bar{w} = \frac{n}{n_T}\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right) + \frac{n}{n_C}\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right),$$

and expanding the square yields

$$\begin{aligned}\mathrm{Var}_\Omega\left[\hat{\tau}\right] = {} & \frac{n_C}{n n_T\left(n-1\right)}\sum_{i=1}^n\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)^2 + \frac{n_T}{n n_C\left(n-1\right)}\sum_{i=1}^n\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right)^2 \\ & + \frac{2}{n\left(n-1\right)}\sum_{i=1}^n\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right).\end{aligned} \tag{11}$$

Now recall that

$$S_n^2(\boldsymbol{y}(1)) = \frac{1}{n-1}\sum_{i=1}^n\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)^2 \text{ and}$$

$$S_n^2(\boldsymbol{y}(0)) = \frac{1}{n-1}\sum_{i=1}^n\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right)^2,$$

which yields

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{n_C}{n n_T}S_n^2(\boldsymbol{y}(1)) + \frac{n_T}{n n_C}S_n^2(\boldsymbol{y}(0)) + \frac{2}{n\left(n-1\right)}\sum_{i=1}^n\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right).$$

Finally, at long last, it follows that

$$
\begin{aligned}
\mathrm{Var}_\Omega\left[\hat{\tau}\right] &= \frac{n_C}{n n_T} S_n^2(\boldsymbol{y}(1)) + \frac{n_T}{n n_C} S_n^2(\boldsymbol{y}(0)) + \frac{2}{n\left(n-1\right)} \sum_{i=1}^{n} \left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1) \right) \left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0) \right) \\
&= \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{y}(1))}{n} - \frac{S_n^2(\boldsymbol{y}(0))}{n} \\
&\quad + \frac{2}{n\left(n-1\right)} \sum_{i=1}^{n} \left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1) \right) \left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0) \right) \\
&= \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} \\
&\quad - \frac{1}{n} \underbrace{\left(S_n^2(\boldsymbol{y}(1)) + S_n^2(\boldsymbol{y}(0)) - \frac{2}{(n-1)} \sum_{i=1}^{n} \left(y_i(1) - \frac{1}{n}\sum_{i=1}^{n} y_i(1) \right) \left(y_i(0) - \frac{1}{n}\sum_{i=1}^{n} y_i(0) \right) \right)}_{= S_n^2(\boldsymbol{\tau})} .
\end{aligned}
$$

As I showed in **Step 3**, the bracketed term in the expression immediately above is equal to $S_n^2(\boldsymbol{\tau})$, which leaves

$$
\frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}. \tag{12}
$$

□

Each of the three terms in this variance has a clear meaning. The estimator $\hat{\tau}$ is the difference between the treated group mean and the control group mean, so its variability comes from two pieces. The first term, $\frac{S_n^2(\boldsymbol{y}(1))}{n_T}$, is the variability that the treated mean contributes. The first term grows with the spread of the treatment potential outcomes, $S_n^2(\boldsymbol{y}(1))$, and shrinks as the number of treated units, n_T, grows. The second term, $\frac{S_n^2(\boldsymbol{y}(0))}{n_C}$, is the analogous variability that the control mean contributes. If the treated and control means came from two *independently* drawn groups, these first two terms would account for the entire variance.

The two means are *not* independent, however, and the third term, $-\frac{S_n^2(\boldsymbol{\tau})}{n}$, corrects for the dependence that complete random assignment induces. The treated and control groups are complementary, since every unit the design sends to treatment is a unit it *withholds* from control, so the treated mean and the control mean are negatively coupled. The coupling runs through the individual treatment effects, which is why the correction involves $S_n^2(\boldsymbol{\tau})$, the finite population variance of the individual effects $\tau_i = y_i(1) - y_i(0)$. Two consequences follow. First, the correction enters with a minus sign and therefore *lowers* the variance; ignoring the correction would overstate the variance, which is what makes the conservative variance estimator (developed in the next section) conservative rather than anti-conservative. Second, when the treatment effect is constant across units, τ_i is the same for all i, so $S_n^2(\boldsymbol{\tau}) = 0$, the correction vanishes, and the spreads of the two sets of potential outcomes fully determine the variance.

The expression I just derived for $\text{Var}_\Omega[\hat{\tau}]$ is the *true* variance of the Difference-in-Means estimator. That expression is mathematically equivalent to Equation 3.4 in Gerber and Green (2012, 57), which differs from the expression in Equation (12) above because the variances and covariance of treated and control potential outcomes in Gerber and Green (2012, Equation 3.4, 57) use a denominator of n as opposed to $n-1$ as in Equations (3), (4) and (5) above. The corollary below establishes this equivalence.

Corollary 1. *An equivalent expression for the finite sample variance of the Difference-in-Means estimator under complete random assignment is*

$$\text{Var}_\Omega\left[\hat{\tau}\right] = \frac{1}{n-1}\left(\frac{n_C \sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{n_T \sigma_n^2(\boldsymbol{y}(0))}{n_C} + 2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))\right), \tag{13}$$

where

$$\sigma_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n}\right)\sum_{i=1}^n \left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)^2 \tag{14}$$

$$\sigma_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n}\right)\sum_{i=1}^n \left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right)^2 \tag{15}$$

$$\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) = \left(\frac{1}{n}\right)\sum_{i=1}^n \left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right)\left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right). \tag{16}$$

Proof. Recall the expression for $\text{Var}_\Omega[\hat{\tau}]$ in Equation (11):

$$\begin{aligned}\text{Var}_\Omega\left[\hat{\tau}\right] = &\frac{n_C}{n n_T\,(n-1)}\sum_{i=1}^n \left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)^2 + \frac{n_T}{n n_C\,(n-1)}\sum_{i=1}^n \left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right)^2 \\ &+ \frac{2}{n\,(n-1)}\sum_{i=1}^n \left(y_i(1) - \frac{1}{n}\sum_{i=1}^n y_i(1)\right)\left(y_i(0) - \frac{1}{n}\sum_{i=1}^n y_i(0)\right).\end{aligned}$$

Then the definitions of $\sigma_n^2(\boldsymbol{y}(1))$, $\sigma_n^2(\boldsymbol{y}(0))$ and $\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))$ above yield

$$\text{Var}_\Omega\left[\hat{\tau}\right] = \frac{n_C}{n_T\,(n-1)}\sigma_n^2(\boldsymbol{y}(1)) + \frac{n_T}{n_C\,(n-1)}\sigma_n^2(\boldsymbol{y}(0)) + \frac{2}{(n-1)}\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))$$

$$= \frac{1}{n-1}\left(\frac{n_C\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{n_T\sigma_n^2(\boldsymbol{y}(0))}{n_C} + 2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))\right),$$

which completes the proof. □

2 Conservative estimation of the variance

Having derived the true variance $\mathrm{Var}_\Omega[\hat{\tau}]$, I now turn to estimating it from the observed data. Absent additional assumptions, we *cannot* estimate $\mathrm{Var}_\Omega[\hat{\tau}]$ unbiasedly. The obstruction is the same in both equivalent expressions for the variance. Each expression depends on a quantity that couples a unit's two potential outcomes, namely the covariance $\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))$ in the Gerber–Green form or the variance of the individual effects, $S_n^2(\boldsymbol{\tau})$, in the Imbens–Rubin form, and we never observe both potential outcomes for any single unit (the fundamental problem of causal inference). What we *can* do is construct an estimable quantity that is at least as large as the true variance and estimate that quantity instead. The resulting estimator is *conservative*, in the sense that on average it weakly overstates the variance, which yields valid confidence intervals (possibly too wide) rather than overconfident ones.

I establish the upper bound in two ways, starting from each of the two equivalent expressions for the variance. Both routes arrive at the *same* conservative estimator. I first record the two expressions for reference. The Imbens–Rubin form (Proposition 1) is

$$\mathrm{Var}_\Omega[\hat{\tau}] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n}, \tag{17}$$

and the Gerber–Green form (the Corollary above) is

$$\mathrm{Var}_\Omega[\hat{\tau}] = \frac{1}{n-1}\left(\frac{n_C\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{n_T\sigma_n^2(\boldsymbol{y}(0))}{n_C} + 2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))\right). \tag{18}$$

2.1 Approach 1 (Gerber–Green expression): bounding the covariance

In Equation (18), we can estimate the variances $\sigma_n^2(\boldsymbol{y}(1))$ and $\sigma_n^2(\boldsymbol{y}(0))$ unbiasedly, but we cannot estimate the covariance $\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))$. The first approach replaces the unestimable covariance with an estimable upper bound. The Cauchy–Schwarz inequality implies that

$$\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) \le \sqrt{\sigma_n^2(\boldsymbol{y}(0))\sigma_n^2(\boldsymbol{y}(1))},$$

and the arithmetic mean–geometric mean (AM–GM) inequality further implies that

$$\sqrt{\sigma_n^2(\boldsymbol{y}(0))\sigma_n^2(\boldsymbol{y}(1))} \leq \frac{\sigma_n^2(\boldsymbol{y}(0)) + \sigma_n^2(\boldsymbol{y}(1))}{2}.$$

Hence $2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) \leq \sigma_n^2(\boldsymbol{y}(0)) + \sigma_n^2(\boldsymbol{y}(1))$. Substituting this bound for the covariance term in Equation (18), and then collecting the two terms in each potential outcome's variance, yields a simple upper bound. The key simplification is that pairing the cross term with the matching squared term gives, for example,

$$\frac{n_C\, \sigma_n^2(\boldsymbol{y}(1))}{n_T} + \sigma_n^2(\boldsymbol{y}(1)) = \frac{n_C + n_T}{n_T}\, \sigma_n^2(\boldsymbol{y}(1)) = \frac{n\, \sigma_n^2(\boldsymbol{y}(1))}{n_T},$$

and likewise for the control variance. Carrying this out,

$$\begin{aligned}
\mathrm{Var}_\Omega\left[\hat{\tau}\right] &= \frac{1}{n-1}\left(\frac{n_C\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{n_T\sigma_n^2(\boldsymbol{y}(0))}{n_C} + 2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1))\right) \\
&\leq \frac{1}{n-1}\left(\frac{n_C\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{n_T\sigma_n^2(\boldsymbol{y}(0))}{n_C} + \sigma_n^2(\boldsymbol{y}(1)) + \sigma_n^2(\boldsymbol{y}(0))\right) \\
&= \frac{1}{n-1}\left(\underbrace{\frac{n_C\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \sigma_n^2(\boldsymbol{y}(1))}_{= n\sigma_n^2(\boldsymbol{y}(1))/n_T} + \underbrace{\frac{n_T\sigma_n^2(\boldsymbol{y}(0))}{n_C} + \sigma_n^2(\boldsymbol{y}(0))}_{= n\sigma_n^2(\boldsymbol{y}(0))/n_C}\right) \\
&= \frac{n}{n-1}\left(\frac{\sigma_n^2(\boldsymbol{y}(1))}{n_T} + \frac{\sigma_n^2(\boldsymbol{y}(0))}{n_C}\right).
\end{aligned}$$

We can estimate the two remaining unknowns, $\sigma_n^2(\boldsymbol{y}(1))$ and $\sigma_n^2(\boldsymbol{y}(0))$, unbiasedly. Following Cochran (1977, Theorem 2.4), unbiased estimators are

$$\begin{aligned}
\widehat{\sigma}_n^2(\boldsymbol{y}(1)) &= \left(\frac{n-1}{n\left(n_T-1\right)}\right)\sum_{i=1}^{n} Z_i\left(Y_i - \frac{1}{n_T}\sum_{i=1}^{n} Z_iY_i\right)^2 \\
\widehat{\sigma}_n^2(\boldsymbol{y}(0)) &= \left(\frac{n-1}{n\left(n_C-1\right)}\right)\sum_{i=1}^{n}(1-Z_i)\left(Y_i - \frac{1}{n_C}\sum_{i=1}^{n}(1-Z_i)Y_i\right)^2.
\end{aligned}$$

These estimators depend only on the *observed* outcomes. The factor Z_i selects the treated units, for which $Y_i = y_i(1)$, and $(1-Z_i)$ selects the control units, for which $Y_i = y_i(0)$. The leading factor $\frac{n-1}{n}$ converts the ordinary sample variance, which is unbiased for $S_n^2(\cdot)$, into an unbiased estimator of $\sigma_n^2(\cdot)$, since $\sigma_n^2(\cdot) = \frac{n-1}{n}S_n^2(\cdot)$. Substituting these into the upper bound gives the conservative variance estimator

$$\widehat{\mathrm{Var}}\,[\hat{\tau}] = \frac{n}{n-1}\left(\frac{\widehat{\sigma}_n^2(\boldsymbol{y}(1))}{n_T} + \frac{\widehat{\sigma}_n^2(\boldsymbol{y}(0))}{n_C}\right). \tag{19}$$

2.2 Approach 2 (Imbens–Rubin expression): dropping the variance of the individual effects

The second approach starts from Equation (17) and is more direct. The only term we cannot estimate is $S_n^2(\boldsymbol{\tau})$, the finite population variance of the individual effects, since computing $S_n^2(\boldsymbol{\tau})$ would require knowing both potential outcomes for each unit. Notice that $S_n^2(\boldsymbol{\tau})$ enters with a *minus* sign and is always nonnegative. Therefore

$$\mathrm{Var}_\Omega\,[\hat{\tau}] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \underbrace{\frac{S_n^2(\boldsymbol{\tau})}{n}}_{\geq 0} \leq \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C}.$$

That is, simply *dropping* the nonnegative term we cannot estimate produces an upper bound. The two remaining terms depend on one potential outcome per unit, and the corresponding sample variances within each group estimate them unbiasedly,

$$\widehat{S}_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n_T-1}\right)\sum_{i=1}^n Z_i\left(Y_i - \frac{1}{n_T}\sum_{i=1}^n Z_iY_i\right)^2$$

$$\widehat{S}_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n_C-1}\right)\sum_{i=1}^n (1-Z_i)\left(Y_i - \frac{1}{n_C}\sum_{i=1}^n (1-Z_i)Y_i\right)^2,$$

which are well defined because $n_T \geq 2$ and $n_C \geq 2$. Substituting these into the upper bound gives the conservative variance estimator

$$\widehat{\mathrm{Var}}\,[\hat{\tau}] = \frac{1}{n_T}\widehat{S}_n^2(\boldsymbol{y}(1)) + \frac{1}{n_C}\widehat{S}_n^2(\boldsymbol{y}(0)). \tag{20}$$

2.3 The two approaches give the same estimator, and how conservative it is

The two routes produce the *same* estimator. To see the equivalence, recall that $\widehat{\sigma}_n^2(\boldsymbol{y}(1)) = \frac{n-1}{n}\widehat{S}_n^2(\boldsymbol{y}(1))$ (compare the two pairs of estimators above), so that

$$\frac{n}{n-1}\frac{\widehat{\sigma}_n^2(\boldsymbol{y}(1))}{n_T} = \frac{n}{n-1}\cdot\frac{n-1}{n}\cdot\frac{\widehat{S}_n^2(\boldsymbol{y}(1))}{n_T} = \frac{\widehat{S}_n^2(\boldsymbol{y}(1))}{n_T},$$

and likewise for the control term. Hence Equation (19) and Equation (20) are identical; bounding the covariance via the Cauchy–Schwarz/AM–GM route and dropping the nonnegative variance of the individual effects are two ways of describing the same approximation.

Finally, the conservative estimator overstates the variance by an amount we can derive exactly. Because $\widehat{S}_n^2(\boldsymbol{y}(1))$ and $\widehat{S}_n^2(\boldsymbol{y}(0))$ are unbiased for $S_n^2(\boldsymbol{y}(1))$ and $S_n^2(\boldsymbol{y}(0))$,

$$\mathrm{E}_\Omega\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} = \mathrm{Var}_\Omega\left[\hat{\tau}\right] + \frac{S_n^2(\boldsymbol{\tau})}{n},$$

where the last equality uses Equation (17). Therefore

$$\mathrm{E}_\Omega\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right]\right] - \mathrm{Var}_\Omega\left[\hat{\tau}\right] = \frac{S_n^2(\boldsymbol{\tau})}{n} \geq 0,$$

so the estimator's upward bias is exactly $S_n^2(\boldsymbol{\tau})/n$. The estimator is conservative in general, and it is *exactly* unbiased when the individual treatment effects are constant across units, i.e., when $S_n^2(\boldsymbol{\tau}) = 0$ (equivalently, when $2\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) = \sigma_n^2(\boldsymbol{y}(0)) + \sigma_n^2(\boldsymbol{y}(1))$, so that the Cauchy–Schwarz/AM–GM bound in Approach 1 holds with equality).

3 Extension: variance and its estimation under simple random assignment

So far the assignment mechanism has been complete random assignment, with the counts fixed at n_T and n_C by design. Under *simple random assignment*, each unit receives treatment independently with a common assignment probability, the two degenerate assignments (all treated and all control) are excluded, and the count $n_T(\boldsymbol{Z})$ is a nondegenerate random variable with support $\{1, \ldots, n-1\}$; the setup and unbiasedness guides develop this mechanism in full. The estimator is the ratio form noted in the setup, with the realized counts $n_T(\boldsymbol{Z})$ and $n_C(\boldsymbol{Z})$ in the denominators.

The conditioning argument that the unbiasedness guide uses for the expectation extends to the variance, for one reason: Conditional on the event $\{n_T(\boldsymbol{Z}) = n_T\}$, the distribution of $\boldsymbol{Z}$ *is* the complete random assignment distribution, as the unbiasedness guide proves (see also Pashley et al., 2021), so every result above holds conditionally, word for word. In particular,

$$\mathrm{Var}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z}) = n_T\right] = \frac{S_n^2(\boldsymbol{y}(1))}{n_T} + \frac{S_n^2(\boldsymbol{y}(0))}{n_C} - \frac{S_n^2(\boldsymbol{\tau})}{n},$$

which is Equation (12) evaluated at the realized counts. Two facts then turn conditional statements into unconditional ones. First, the conditional expectation of the estimator is the *same* constant at every count: The unbiasedness guide proves $\mathrm{E}_\Omega[\hat{\tau} \mid n_T(\boldsymbol{Z}) = k] = \tau_{\mathrm{SATE}}$ for every $k \in \{1, \ldots, n-1\}$. Second, the conservative estimator's conditional overshoot, $S_n^2(\boldsymbol{\tau})/n$, also does not depend on the count. Averaging over the count therefore costs nothing, as the following proposition records.

Proposition 2. *Under simple random assignment,*

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = S_n^2(\boldsymbol{y}(1))\ \mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] + S_n^2(\boldsymbol{y}(0))\ \mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] - \frac{S_n^2(\boldsymbol{\tau})}{n}.$$

Proof. By the law of total variance, conditioning on the count,

$$\mathrm{Var}_\Omega\left[\hat{\tau}\right] = \mathrm{E}_\Omega\Big[\mathrm{Var}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z})\right]\Big] + \mathrm{Var}_\Omega\Big[\mathrm{E}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z})\right]\Big].$$

The second term is zero, because $\mathrm{E}_\Omega[\hat{\tau} \mid n_T(\boldsymbol{Z}) = k] = \tau_{\mathrm{SATE}}$ for every k, and a constant has no variance. For the first term, conditional on $n_T(\boldsymbol{Z}) = k$ the distribution of $\boldsymbol{Z}$ is complete random assignment with k treated units, so Proposition 1 applies with $n_T = k$ and $n_C = n - k$:

$$\mathrm{Var}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z}) = k\right] = \frac{S_n^2(\boldsymbol{y}(1))}{k} + \frac{S_n^2(\boldsymbol{y}(0))}{n-k} - \frac{S_n^2(\boldsymbol{\tau})}{n}.$$

The conditional variance is a random variable that depends on $\boldsymbol{Z}$ only through its count, so the outer expectation averages the display above over the distribution of the count of treated units:

$$\mathrm{E}_\Omega\Big[\mathrm{Var}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z})\right]\Big] = \sum_{k=1}^{n-1} \mathrm{Pr}_\Omega\left(n_T(\boldsymbol{Z}) = k\right)\left(\frac{S_n^2(\boldsymbol{y}(1))}{k} + \frac{S_n^2(\boldsymbol{y}(0))}{n-k} - \frac{S_n^2(\boldsymbol{\tau})}{n}\right).$$

The three variances are fixed quantities, so the linearity of expectations splits the sum into three pieces and pulls each variance out front:

$$\begin{aligned}\mathrm{E}_\Omega\Big[\mathrm{Var}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z})\right]\Big] &= S_n^2(\boldsymbol{y}(1))\sum_{k=1}^{n-1}\frac{\mathrm{Pr}_\Omega\left(n_T(\boldsymbol{Z}) = k\right)}{k} + S_n^2(\boldsymbol{y}(0))\sum_{k=1}^{n-1}\frac{\mathrm{Pr}_\Omega\left(n_T(\boldsymbol{Z}) = k\right)}{n-k} \\ &\quad - \frac{S_n^2(\boldsymbol{\tau})}{n}\sum_{k=1}^{n-1}\mathrm{Pr}_\Omega\left(n_T(\boldsymbol{Z}) = k\right).\end{aligned}$$

The first sum is, by definition, the expected reciprocal $\mathrm{E}_\Omega[1/n_T(\boldsymbol{Z})]$. The second sum is $\mathrm{E}_\Omega[1/n_C(\boldsymbol{Z})]$, because $n_C(\boldsymbol{Z}) = n - n_T(\boldsymbol{Z})$, so the event $\{n_T(\boldsymbol{Z}) = k\}$ is the event $\{n_C(\boldsymbol{Z}) = n - k\}$. The third sum equals one, because the count takes some value in $\{1, \ldots, n-1\}$ with probability one. Substituting the three pieces yields the claim. □

Compared with Equation (12), the only change is that the fixed reciprocals $1/n_T$ and $1/n_C$ become the expected reciprocals $\mathrm{E}_\Omega[1/n_T(\boldsymbol{Z})]$ and $\mathrm{E}_\Omega[1/n_C(\boldsymbol{Z})]$; the correction term $-S_n^2(\boldsymbol{\tau})/n$

is untouched. The expected reciprocals have no simple closed form, but the unbiasedness guide derives the distribution of the count of treated units, $\Pr_\Omega(n_T(\boldsymbol{Z}) = k) = \binom{n}{k} p^k (1-p)^{n-k}/c$ with $c = 1 - p^n - (1-p)^n$, so direct summation computes them exactly:

$$\mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] = \sum_{k=1}^{n-1} \frac{1}{k} \Pr_\Omega\left(n_T(\boldsymbol{Z}) = k\right),$$

and likewise for $\mathrm{E}_\Omega[1/n_C(\boldsymbol{Z})]$, with $\frac{1}{n-k}$ in place of $\frac{1}{k}$. Simulation gives the same numbers without the formula: Draw many assignment vectors from the mechanism, discard any degenerate draws, and average the realized values of $1/n_T(\boldsymbol{Z})$; by the law of large numbers the average converges to the expected reciprocal.

Which mechanism delivers the more precise estimator? The two variance formulas share the correction term $-S_n^2(\boldsymbol{\tau})/n$, so the comparison turns entirely on the reciprocals of the counts — and reciprocals are convex, so variability in the group sizes can only inflate their expected values. Fixing balanced counts therefore weakly improves precision, as the following corollary records for the natural benchmark.

Corollary 2. *Let n be even, let simple random assignment use $p = \frac{1}{2}$, and let complete random assignment fix the balanced counts $n_T = n_C = \frac{n}{2}$. Then the variance of $\hat{\tau}$ under complete random assignment is no larger than under simple random assignment, and it is strictly smaller unless $S_n^2(\boldsymbol{y}(1)) = S_n^2(\boldsymbol{y}(0)) = 0$.*

Proof. The two variances share the term $-S_n^2(\boldsymbol{\tau})/n$, so subtracting the variance in Equation (12), evaluated at $n_T = n_C = \frac{n}{2}$, from the variance in Proposition 2 leaves

$$S_n^2(\boldsymbol{y}(1))\left(\mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] - \frac{2}{n}\right) + S_n^2(\boldsymbol{y}(0))\left(\mathrm{E}_\Omega\left[\frac{1}{n_C(\boldsymbol{Z})}\right] - \frac{2}{n}\right).$$

Under $p = \frac{1}{2}$ the distribution of the count is symmetric, $\Pr_\Omega(n_T(\boldsymbol{Z}) = k) = \Pr_\Omega(n_T(\boldsymbol{Z}) = n-k)$, so $\mathrm{E}_\Omega[n_T(\boldsymbol{Z})] = \frac{n}{2}$. Jensen's inequality for the strictly convex function $x \mapsto \frac{1}{x}$ on the positive reals then gives

$$\mathrm{E}_\Omega\left[\frac{1}{n_T(\boldsymbol{Z})}\right] > \frac{1}{\mathrm{E}_\Omega\left[n_T(\boldsymbol{Z})\right]} = \frac{2}{n},$$

with strict inequality because $n_T(\boldsymbol{Z})$ is nondegenerate, and the same holds for $n_C(\boldsymbol{Z}) = n - n_T(\boldsymbol{Z})$. Both bracketed differences are therefore strictly positive, so the displayed difference of variances is nonnegative, and it is zero exactly when both $S_n^2(\boldsymbol{y}(1))$ and $S_n^2(\boldsymbol{y}(0))$ are zero. □

With any outcome variation at all, the coin-flip design pays a real price for its random group sizes, and the price is exactly the gap between the expected reciprocals and the reciprocals of the expected counts. The same comparison applies whenever the fixed counts of a complete random assignment match the expected counts of a simple random assignment, provided those expected counts are whole numbers. A comparison across mismatched designs can go either way: An extremely unbalanced fixed design can be worse than a balanced coin flip.

The conservative estimator extends in the same way. With random counts the estimator uses the realized counts,

$$\widehat{\mathrm{Var}}\left[\hat{\tau}\right] = \frac{\widehat{S}_n^2(\boldsymbol{y}(1))}{n_T(\boldsymbol{Z})} + \frac{\widehat{S}_n^2(\boldsymbol{y}(0))}{n_C(\boldsymbol{Z})},$$

and each arm's sample variance requires at least two units, for the same reason the complete random assignment analysis assumes $n_T, n_C \geq 2$. Condition therefore on the event $A := \{2 \leq n_T(\boldsymbol{Z}) \leq n-2\}$. Conditional on any count in A, the previous section applies verbatim, so the overshoot is exactly $S_n^2(\boldsymbol{\tau})/n$ at every such count, and averaging over the counts in A preserves it, again because the conditional expectation of $\hat{\tau}$ does not vary with the count:

$$\mathrm{E}_\Omega\left[\widehat{\mathrm{Var}}\left[\hat{\tau}\right] \mid A\right] - \mathrm{Var}_\Omega\left[\hat{\tau} \mid A\right] = \frac{S_n^2(\boldsymbol{\tau})}{n}.$$

The estimator is exactly as conservative under simple random assignment as under complete random assignment.

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